



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 4 • STANDARD 6.AT.4

## Determine the Whole Given the Part and Percent

Lesson 4-5 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can solve problems involving discounts, markups, tax, and tips using percents.

**Language Objective:** I can explain my solution using the words discount, markup, tax, tip, and percent.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Percent Problem Solving

Discount

Markup

Tax

Tip

Percent



Tap any word to see what it means and a picture.

1

— Use a percent to find a part, a whole, or a percent change in a real situation.

2

An amount taken off the original price is a .

3

The percent a store adds to its cost to set a higher selling price is a

.

4

The percent added to a purchase that goes to the government is

.

5

The percent of a bill you add for good service is a .

6

A number out of 100 is a .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

JUSTIFY

**Is your friend correct? Show your work to prove it?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Percent Problem Solving** — Use a percent to find a part, a whole, or a percent change in a real situation.  
*Resolución de problemas con porcentajes — Usar un porcentaje para hallar una parte, un total o un cambio porcentual en una situación real.*

☐ **Discount** — Money taken off the first price to make it cheaper.  
*Descuento — Dinero que se quita del precio original para que sea más barato.*

☐ **Markup** — The extra amount a store adds to what it paid, found by taking a percent of that cost.  $\text{Cost} + \text{markup} = \text{selling price}$ .  
*Recargo (margen) — La cantidad extra que una tienda suma a lo que pagó, hallada tomando un porcentaje de ese costo.  $\text{Costo} + \text{recargo} = \text{precio de venta}$ .*

☐ **Tax** — Extra money added to the price of a purchase for the government, found by taking a percent of the price.  $\text{Price} + \text{tax} = \text{total}$ .  
*Impuesto — Dinero extra que se suma al precio de una compra para el gobierno, hallado tomando un porcentaje del precio.  $\text{Precio} + \text{impuesto} = \text{total}$ .*

☐ **Tip** — Extra money you add for good service, found by taking a percent of the bill.  $\text{Bill} + \text{tip} = \text{total}$ .  
*Propina — Dinero extra que agregas por un buen servicio, hallado tomando un porcentaje de la cuenta.  $\text{Cuenta} + \text{propina} = \text{total}$ .*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**First I find the discount: 30% of \$50 = \$\_\_\_\_. The sale price is \$50 - \$\_\_\_\_ = \$\_\_\_\_. Then I add 6% tax: \$\_\_\_\_  $\times$  0.06 = \$\_\_\_\_. The total is \$\_\_\_\_ + \$\_\_\_\_ = \$\_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** A discount lowers the price, while tax, tip, and markup raise it.

**Evidence:** Students sort discount as lowering price and tax/tip/markup as raising it, and recognize all are a percent of a price.

**Reasoning:** This evidence connects the definition of percent to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **First I find the discount:**  $30\%$  of  $\$50 = \$$  \_\_\_\_ . The sale price is  $\$50 - \$$  \_\_\_\_ =  $\$$  \_\_\_\_ .  
Then I add **6% tax:**  $\$$  \_\_\_\_  $\times 0.06 = \$$  \_\_\_\_ . The total is  $\$$  \_\_\_\_ +  $\$$  \_\_\_\_ =  $\$$  \_\_\_\_ .
- **My answer is** \_\_\_\_ .

*Mi respuesta es* \_\_\_\_ .

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is** \_\_\_\_ .  
*Mi afirmación es* \_\_\_\_ .
- **I know because** \_\_\_\_ .

*Lo sé porque* \_\_\_\_ .

- **So** \_\_\_\_ .

*Entonces* \_\_\_\_ .

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
  - **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
  - **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*
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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Percent Problem Solving
2. **Notes 2:** Discount
3. **Notes 3:** Markup
4. **Notes 4:** Tax
5. **Notes 5:** Tip
6. **Notes 6:** Percent
7. **Watch for this mistake:** *A common mistake in Use Percent to Solve Problems is combining the discount and tax into one single percent change instead of applying them one after another. For a \$40 item with a 25% discount and 8% sales tax, a student might compute  $40 \times (1 - 0.25 + 0.08) = 40 \times 0.83 = \$33.20$ , treating the tax as if it applies to the original price. But tax is charged on the discounted price, not the original price: first take 25% off \$40 to get  $40 \times 0.75 = \$30$ , then add 8% tax on that \$30:  $30 \times 1.08 = \$32.40$ . The correct final price is \$32.40, not \$33.20 — always compute the discount first, then apply tax to the new lower price.*
8. **Try It 1:** A. \$27 — Tax:  $0.08 \times 25 = 2$ . Total:  $25 + 2 = \$27$ .
9. **Try It 2:** A. \$6 — Tip:  $0.15 \times 40 = \$6$ .